

# Hao Li

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**EDUCATION**      **The University of Manchester, UK**      2025  
*Ph.D. in Computer Science*

**The University of Leeds, UK**      2020  
*M.Sc. in Advanced Computer Science (AI)*

**Northeast Petroleum University, China**      2017  
*B.Sc. in Information and Computing Science*

**RESEARCH DIRECTIONS**      **Foundation Model, Post-training and Alignment, Reinforcement Learning**

**WORK EXPERIENCE**      **ERNIE Team, Baidu**      Jan.2026 – Present  
Researcher, China

- Technical Staff member of the General Capabilities Post-training Group within the ERNIE Foundation Model Team. My role bridges cutting-edge research and large-scale industrial application, focusing on Reward Modeling and Reinforcement Learning (RL) to maintain ERNIE's position as a state-of-the-art LLM.
- **Unified RL Training:** Deep Involved the post-training alignment for flagship models (ERNIE-5.0-0110, ERNIE-4.5-Turbo), designing a holistic strategy to integrate disparate capabilities (Deep Research, Coding, Agent, Creative Writing) into a single general model, successfully mitigating alignment tax.
- **Ultra-Scale Optimization:** Validated and deployed frontier RL algorithms and data mixture recipes on 30B to 2400B parameter models series, achieving state-of-the-art performance benchmarks in production.
- **High-Performance Infrastructure:** Co-architected the distributed RL training infrastructure for Reward Modeling and Policy Optimization, significantly optimizing throughput and stability on massive GPU clusters.

**Imperial College London**      Nov.2025 – May.2026  
*Postdoctoral Researcher, UK*

- Led the research and prototyping of a **differentially private** synthetic data generation system for deployment within Trusted Research Environments (TREs).
- Collaborated with engineering teams to transform research code into containerised, production-ready software integrated into the TREvolution platform.

**Microsoft Research**      Aug.2024 – May.2025  
*Research Internships, China*

- Contributed to a **Foundation Model for Medical Time Series**, extending **Next-Token Prediction** to arbitrary time points via **Neural ODEs**, enhancing modeling of irregularly sampled clinical data.
- Core developer of the open-source project **TimeCraft** (1k+ GitHub stars, responsible for **Controllable Text-to-Time-Series Generation**:

## NOTABLE PROJECT

**TimeCraft** (*Microsoft Research Asia*)  
*Core Developer*

Mar 2024 – Feb 2025

- Developed a **Universal Diffusion-based Framework** for time-series generation, leveraging **Diffusion Models** to achieve state-of-the-art performance in synthetic data synthesis.
- Enabled controllable generation via **Plain Language Descriptions**, allowing users to synthesize complex temporal patterns (e.g., medical trends, market behavior) with strong **cross-domain generalization**.
- Achieved **1.1k+ GitHub Stars** and wide community adoption; the framework serves as a benchmark for high-fidelity time-series generation tasks.

## Awards & Funding

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|------------------------------------------------------------------|------|
| Research Grant (Medical Text Mining with Multi-task Learning)    | 2022 |
| Outstanding Student Scholarship, University of Leeds             | 2019 |
| Outstanding Graduate Scholarship, Northeast Petroleum University | 2017 |
| First-class Scholarship, Northeast Petroleum University          | 2016 |
| National First Prize, China College Big Data Competition         | 2016 |

## PUBLICATIONS Conference Proceedings

1. **Hao Li**, Yizheng Sun, Viktor Schlegel, Kailai Yang, Riza Batista-Navarro, and Goran Nenadic. “[Arg-LLaDA: Argument Summarization via Large Language Diffusion Models and Sufficiency-Aware Refinement](#)”, **Expected to ACL 2026** (*Meta Review:4, Review Score:3, 3.5, 4.5*)
2. **Hao Li**, Bowen Deng, Chang Xu, Zhiyuan Feng, Viktor Schlegel, Yu-Hao Huang, Yizheng Sun, Jingyuan Sun, Kailai Yang, Yiyao Yu, Jiang Bian. “[MIRA: Medical Time Series Foundation Model for Real-World Health Data](#)”, **NeurIPS 2025** (*CCF-A, A\*, 24.5% acceptance rate*)
3. **Hao Li**, Yuhao Huang, Chang Xu, Viktor Schlegel, Renhe Jiang, Riza Batista-Navarro, Goran Nenadic, Jiang Bian. “[BRIDGE: Bootstrapping Text to Control Time-series Generation via Multi-agent Iterative Optimization and Diffusion Modelling](#)”, **ICML 2025** (*CCF-A, A\*, 25.5% acceptance rate*)
4. **Hao Li**, Yuping Wu, Viktor Schlegel, Riza Batista-Navarro, Tharindu Madusanka, Iqra Zahid, Jiayan Zeng, Xiaochi Wang, Xinran He, Yizhi Li, Goran Nenadic. “[Which Side Are You On? A Multi-task Dataset for End-to-End Argument Summarisation and Evaluation](#)”, **Findings of ACL 2024**
5. **Hao Li**, Viktor Schlegel, Riza Batista-Navarro, and Goran Nenadic. “[Do You Hear The People Singing? Key Point Analysis via Iterative Clustering and Abstractive Summarisation](#)”, **ACL 2023** (*CCF-A, A\*, 23.5% acceptance rate*)
6. Bowen Deng, Chang Xu, **Hao Li**, Yu-Hao Huang, Min Hou, Jiang Bian. “[TarDiff: Target-Oriented Diffusion Guidance for Synthetic Electronic Health Record Time Series Generation](#)”, **KDD 2025** (*CCF-A, A\*, 20.1% acceptance rate*)
7. Yizheng Sun, **Hao Li**, Chang Xu, Hongpeng Zhou, Chenghua Lin, Riza Batista-Navarro, Jingyuan Sun. “[Does Acceleration Cause Hidden Instability in Vision Language Models? Uncovering Instance-Level Divergence Through a Large-Scale Empirical Study](#)”, **EMNLP 2025** (*CCF-B, A, 23.3% acceptance rate*)
8. Yizheng Sun, Qinshi Rui, **Hao Li**, Chenghua Lin, Riza Batista-Navarro. “[LVPPruning: An Effective yet Simple Language-Guided Vision Token Pruning Approach for Multi-modal Large Language Models](#)”, **Findings of NAACL 2025**

9. Yizhi Li, **Hao Li**, Chenghua Lin and 17 others. “CIF-Bench: The Challenging Chinese Instruction-Following Benchmark to Measure Generalizability of Large Language Models”, **Findings of ACL 2024**
10. Tharindu Madusanka, Iqra Zahid, **Hao Li**, Ian Pratt-Hartmann and Riza Batista-Navarro. “Not All Quantifiers are Equal: Probing Transformer-based Language Models’ Understanding of Generalised Quantifiers”, **EMNLP 2023 (CCF-B, A, 23.3% acceptance rate)**

## Journal Articles

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11. Yuanyuan Liang, Yanbing Ju, Xiao-Jun Zeng, **Hao Li**, Peiwu Dong, Tian Ju. “A user-generated content-based social network large-scale group decision-making approach in healthcare service: Case study of general practitioners selection in UK”, **Expert Systems with Applications (Q1, SJR 1.875, IS 9.29)**

## Extended Abstracts, Workshops, Posters

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12. **Hao Li**, Yuping Wu, Viktor Schlegel, Riza Batista-Navarro, Thanh-Tung Nguyen, Abhinav Ramesh Kashyap, Xiaojun Zeng, Daniel Beck, Stefan Winkler, Goran Nenadic. “PULSAR: Pre-training with Extracted Healthcare Terms for Summarising Patients’ Problems and Data Augmentation with Black-box Large Language Models”, **BioNLP @ ACL 2023**
13. Poehl.Paul, **Hao Li**, Viktor Schlegel, Anil Bharath. “Generating Realistic Multi-Beat ECG Signals”, **DSP 2025**
14. Aishik Nagar, Viktor Schlegel, Thanh-Tung Nguyen, **Hao Li**, Yuping Wu, Kulhan Binici, Stefan Winkler. “LLMs are not Zeroshot Reasoners for Biomedical Information Extraction”, **Insights @ NAACL 2025**
15. Yizheng Sun, **Hao Li**, Riza Batista-Navarro. “LanViKD: Language-Vision Cross Modal Knowledge Distillation for Egocentric Action Recognition”, **HAII @ ECAI 2024**
16. Viktor Schlegel, **Hao Li**, Yuping Wu, Anand Subramanian, Thanh-Tung Nguyen, Abhinav Ramesh Kashyap, Daniel Beck, Xiaojun Zeng, Riza Theresa Batista-Navarro, Stefan Winkler, Goran Nenadic. “Large Language Models Augmented by Synthetic Dialogue Convert Patient Dialogues to Medical Records”, **Clef 2023 Working Note**

## In Review & Preprint

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17. **Hao Li**, Viktor Schlegel, Riza Batista-Navarro, and Goran Nenadic “Large Language Models in Argument Mining: A Survey”, **Preprint**
18. Zhang Xin, Kailai Yang, **Hao Li**, Chenyue Li, Qiyu Wei, Sophia Ananiadou “Implicit Graph, Explicit Retrieval: Towards Efficient and Interpretable Long-horizon Memory for Large Language Models”, **Preprint**
19. Kailai Yang, Xiao Liu, Lei Ji, **Hao Li**, Yeyun Gong, Peng Cheng, Mao Yang. “Data Mixing Agent: Learning to Re-weight Domains for Continual Pre-training”, **Preprint**
20. Xu Zhang, Junwei Deng, Chang Xu, **Hao Li**, Jiang Bian “Continuous Time Series Generation with Irregular Observations”, **Preprint**

21. Xu Zhang, **Hao Li**, Chang Xu, Yuhao Huang, Jiang Bian “A Survey on Time-Series Generation: Taxonomy, Review and Prospects”, **Preprint**
22. Hui Sun, Chang Xu, Haonan Xie, **Hao Li**, Yu-Hao Huang, Chuheng Zhang, Ming Jin, Xiaoguang Liu, Gang Wang, Jiang Bian “ChatAD: Reasoning-Enhanced Time-Series Anomaly Detection with Multi-Turn Instruction Evolution and Generalization Optimization”, **Preprint**
23. Yidan Sun, Viktor Schlegel, Srinivasan Nandakumar, Iqra Zahid, Yuping Wu, Yulong Wu, **Hao Li**, Jie Zhang, Warren Del-Pinto, Goran Nenadic, Siew Kei Lam, Anil Anthony Bharath. “SYNBENCH: A Benchmark for Differentially Private Text Generation”, **Preprint**
24. Yuping Wu, Viktor Schlegel, Warren Del-Pinto, Srinivasan Nandakumar, Iqra Zahid, Yidan Sun, **Hao Li**, Usama Farghaly Omar, Amirah Jasmine, Arun-Kumar Kaliya-Perumal, Chun Shen Tham, Gabriel Connors, Anil Anthony Bharath, Goran Nenadic. “Privacy-Preserving Generation of Clinical Narratives from Medical Terminologies”, **Preprint**
25. Yuping Wu, **Hao Li**, Hongbo Zhu, Goran Nenadic, Xiao-Jun Zeng. “Unifying Extractive and Abstractive Summarization within the Single Encoder-Decoder Framework”, **Preprint**
26. Hongbo Zhu, **Hao Li**, Angelo Cangelosi. “Representation Understanding via Activation Maximization”, **Preprint**